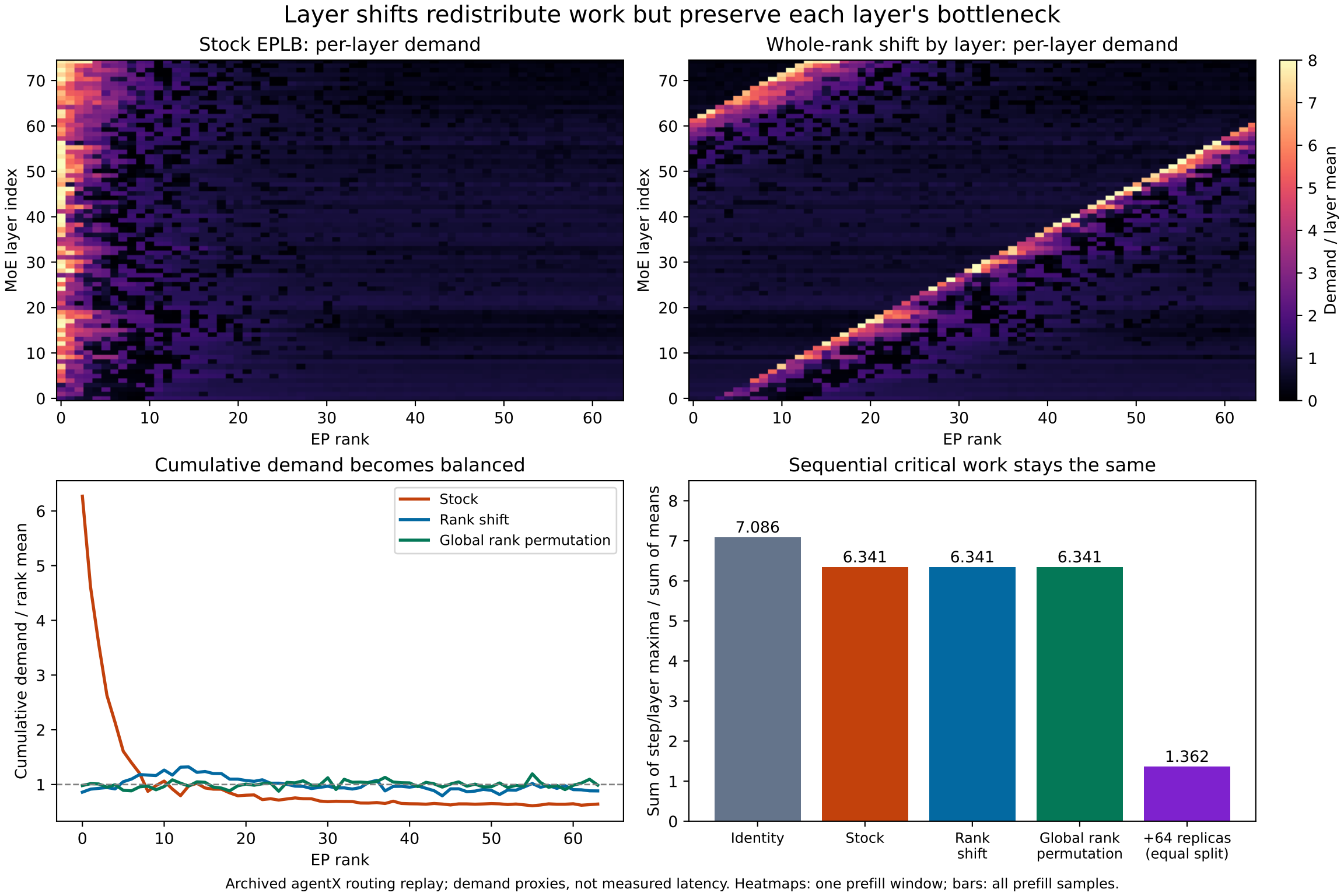
Layer shifts redistribute work but preserve each layer's bottleneck
Stock EPLB: per-layer demand
Whole-rank shift by layer: per-layer demand
MoE layer index
EP rank
Demand / layer mean
Cumulative demand becomes balanced
Cumulative demand / rank mean
Stock
Rank shift
Global rank permutation
Sequential critical work stays the same
Sum of step/layer maxima / sum of means
7.086
6.341
6.341
6.341
1.362
Identity
Stock
Rank shift
Global rank permutation
+64 replicas (equal split)
Archived agentX routing replay; demand proxies, not measured latency. Heatmaps: one prefill window; bars: all prefill samples.